

Basic (design on paper before touching code) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 3 depth levels: **Basic** (the plain idea), **Intermediate** (a real, filled-in prediction), **Advanced** (the failure modes and hidden costs a first guess usually misses). Read Basic first even if you already feel confident about the patterns — it's the fastest way to spot exactly what each deeper level adds.

- [Hint 1 — The idea, and the exact pieces](#)
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Hint 1 — The idea, and the exact pieces

Basic Version

Pick one concrete task, like "research a topic, then write a short summary of it." You're going to design it three separate ways, on paper, without writing a line of code.

For each design, picture how the task would actually flow from start to finish, and who (which agent) does what, in what order.

Before writing anything, just answer for yourself: which of the three designs do you already suspect will be fastest? Which will be most reliable? Write your gut guess down — you'll come back to it.

Things to use:

- One concrete task, written down in one sentence.
- This document's Core Concepts "Every pattern" table — specifically the Single agent, Sequential, and Supervisor rows.
- One paragraph per design, describing the flow in plain words.
- One gut-guess prediction, written down before you reason it out carefully.

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Intermediate Version

The three designs to sketch, using this document's Core Concepts table as your reference for what each one actually means:

- **Single agent** — one agent, one system prompt, with both a "research" tool and a "write" ability. No handoffs at all.
- **Sequential** — two agents (or two clearly separated steps), research always running first, its output feeding directly into the write step, in a fixed order.
- **Supervisor** — a router agent that decides which specialist to call, in what order, based on the task — even though for this simple task the order is probably still research-then-write, the supervisor decides that dynamically rather than it being hardcoded.

For each, write one paragraph describing the flow, then one line predicting: cost (roughly how many LLM calls), speed (roughly how much wall-clock delay), and reliability (what's most likely to go wrong).

The exact pieces:

- **Cost** — single agent: 1-2 calls (it might call its research tool internally, then answer). Sequential: 2 calls minimum, one per stage. Supervisor: 2 calls for the work, plus 1+ extra calls for the routing decision itself — the supervisor's own reasoning about who to call isn't free.
- **Speed** — single agent and sequential both run everything one after another (no parallelism here, since write genuinely needs research's output first). Supervisor adds the routing decision's own delay on top of sequential's delay.
- **Reliability** — single agent: fewer moving parts, but a single system prompt trying to be good at two different jobs can do both a little worse. Sequential: very predictable, since each stage has one clear job — but a bad research result guarantees a bad write result, with nothing to catch it. Supervisor: adds a new failure mode entirely — the supervisor routing to the wrong specialist, or looping between them unnecessarily.

Write your three paragraphs and three one-line predictions before Hint 2.

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Advanced Version

A first-pass guess usually undercounts two things: hidden costs, and how a design fails, not just whether it "works."

On hidden costs — the supervisor's routing decision is easy to write off as "basically free," since it's not doing the actual research or writing work. It isn't free: it's a real LLM call, reasoning over the current state, and for a task this small it can easily cost as much as one of the two real work stages. Write down, honestly, whether your supervisor prediction accounted for that, or quietly assumed routing was instant and free.

On failure modes — don't just ask "does this design work?", ask "what does it do when something goes wrong partway through?" A single agent that mishandles its research tool call still produces *something*, confidently, with nothing external to catch it. Sequential's research stage failing propagates straight through to write, guaranteed, since nothing checks the handoff. Supervisor adds a failure mode neither of the other two has at all: routing to the wrong specialist, or calling the same one twice because its own state-tracking is off.

The real design question isn't just "which design is fastest on paper" — it's "which design's most likely failure would you actually notice, and which would quietly produce a confident-sounding wrong answer?"

The extra pieces needed:

- A one-line "if this goes wrong, here's what happens" note added under each of your three paragraphs, not just the happy-path flow.
- An honest accounting of the supervisor's routing call as a real, separately-counted cost — not folded silently into "2 calls for the work."
- The out-loud defense this document's "How to Practice" section asks for: could you explain to a skeptical senior engineer, in one sentence, why the task you picked would or wouldn't justify 3 agents over 1?

Sketch this hardened version yourself before checking Hint 2.

Difference between Basic, Intermediate, and Advanced: Basic names the 3 designs and asks for a plain-language paragraph and a gut guess per design. Intermediate turns that gut guess into a structured, defensible prediction across cost/speed/reliability, using the document's own pattern table as the reference. Advanced asks what a first guess usually misses — the routing call's own real cost, and each design's specific failure mode, not just its happy path — which is the difference between a guess you can be proud of and one that survives being questioned by someone skeptical.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
task: research a topic and write a short summary

design 1 -- single agent:
  one agent, has a research tool and just writes the summary itself
  guess: cheapest, fastest, but summary quality depends on one prompt doing two

design 2 -- sequential:
  agent A researches, hands text to agent B, agent B writes
  guess: a bit more cost (2 calls), same speed order, more reliable since each

design 3 -- supervisor:
  a router decides to call research agent, then write agent
  guess: most cost (routing + 2 calls), slower (extra routing step), same reliable
```

Expected output if you "run" just this: nothing runs — this whole exercise is paper only. What you have here is a 3-way comparison sketch. The next exercises (`sequential_measure` , `supervisor_compare`) are where you'll actually build 2 of these 3 and find out if the guess above was right.

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Intermediate Version

task: "research the top 3 causes of X, then write a 3-sentence summary"

design 1 -- single agent

flow: one agent gets the task, calls a research tool as needed, then writes the summary itself in the same turn

cost: ~1-2 LLM calls

speed: fastest -- no handoff overhead

reliability risk: the single system prompt has to be good at both research judgment and writing style at once

design 2 -- sequential

flow: research_agent(topic) -> summary; write_agent(summary) -> final text

cost: ~2 LLM calls

speed: same order as design 1, roughly, plus a small handoff cost

reliability risk: no reliability improvement over research quality itself -- if research is bad, the write step faithfully writes a bad summary

design 3 -- supervisor

flow: supervisor reads the task, decides to call research_agent, gets result, decides to call write_agent, gets result, returns

cost: ~3+ LLM calls (2 for the work, 1+ for routing)

speed: slowest of the three -- routing decisions add real delay

reliability risk: everything sequential has, plus the supervisor could route to the wrong specialist or loop unnecessarily

Fill in your own chosen task the same way — one paragraph and one cost/speed/reliability line per design — before moving to Advanced.

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Advanced Version

Here's almost the whole write-up for a different task ("plan a trip"), with the failure-mode line Advanced Hint 1 asked for — fill in your own task the same way, don't copy this one directly:

```
task: "plan a 3-day trip: find flights, then find a hotel near the arrival airport"

single agent (one agent, two tools):
  cost: ~1-2 calls -- the agent can call both tools in one turn if it reasons w
  speed: fastest -- no agent-to-agent handoff delay
  reliability risk: it might search for a hotel before it has flight dates, if
  own internal reasoning doesn't enforce the order -- nothing external enforces
  if it goes wrong: a confident-sounding hotel suggestion with no real flight
  dates behind it -- nothing flags this, since the same agent "trusts itself"

sequential (flight_agent -> hotel_agent, fixed order):
  cost: ~2 calls, one per stage
  speed: medium -- hotel search genuinely can't start until flight dates are
  known, so there's no real parallelism to lose here anyway
  reliability risk: low for ordering (it's enforced by code, not the model's
  judgment), but a flight_agent mistake still propagates straight through
  if it goes wrong: a bad flight pick guarantees a hotel search built on it --
  the failure is very visible in the flight step, at least, since nothing hides

supervisor (router decides who runs next):
  cost: highest -- the routing decision itself is 1+ extra calls, real and
  separately counted, not folded into "2 calls for the work"
  speed: slowest -- same work as sequential, plus routing overhead
  reliability risk: everything sequential has, plus a new one: the router could
  send the task to hotel_agent before flight_agent has run
  if it goes wrong: the router silently loops flight_agent twice, or sends
  hotel_agent an empty flight-dates field it doesn't know how to handle
```

What's missing: your own task, filled in the same way including the "if it goes wrong" line, plus the follow-up commitment to actually build the sequential and supervisor versions in the next two exercises and compare against these guesses. Do that, then check the [Solution](#).

Difference between Basic, Intermediate, and Advanced: Basic gives you the pseudocode shape and a filled example to imitate loosely. Intermediate fills that shape in with real reasoning per design, at the same depth you should aim for on your own task. Advanced adds the piece a first guess almost always skips — a specific "if this goes wrong" line for each design, and the routing call

counted as its own real cost rather than hand-waved away — which is what turns a guess into something you could actually defend out loud.

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Full solution: [Show me the solution](#)